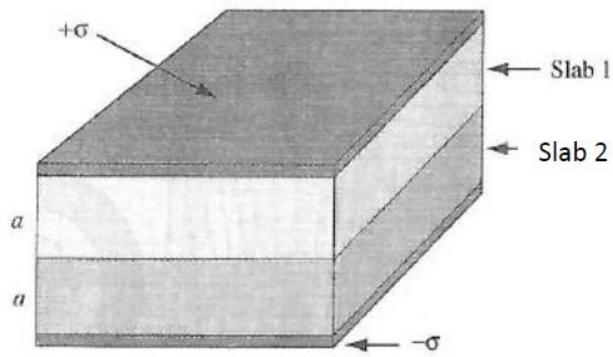
Tutorial – 10

Department of Physics **Shiv Nadar University**

---

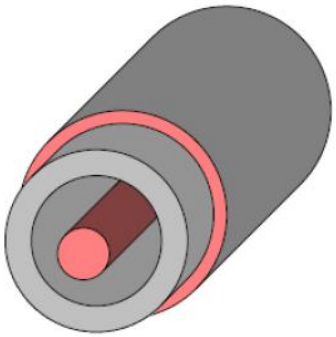
1. The space between the ***metallic*** plates of a parallel plate capacitor is filled with two slabs of linear dielectric material. Each slab has thickness 'a', so the total distance between the plates is '2a'. Slab 1 (top) has the dielectric constant of 2, and slab 2 (bottom) has a dielectric constant of 1.5. The free charge density on the top plate is σ and on the bottom plate $-\sigma$.

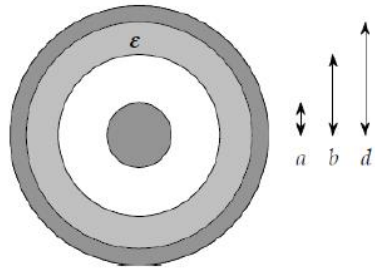
(a) Find the electric displacement ***D*** in each slab. [Note that inside a metal, **D = 0**]. (b) Find electric field ***E*** in each slab. (iii) Find the polarization ***P*** in each slab. (iv) Find the potential difference between the plates. (v) Find the location and amount of all bound charge. (vi) Now that you know all the charge (free and bound), recalculate the field in each slab, and confirm your answer to (b).



2. A sphere of linear dielectric material has embedded in it a uniform ***free*** charge density ρ. Find the potential at the center of the sphere (relative to infinity), if its radius is ***R*** and the **dielectric constant** is ϵ_r.

3. A certain coaxial cable consists of a copper wire, radius a, surrounded by a concentric copper tube of inner radius 'c' (see figure below). The space between is partially filled (from 'b' out to 'c') with material of dielectric constant ϵ_r. Find the capacitance per unit length of this cable.

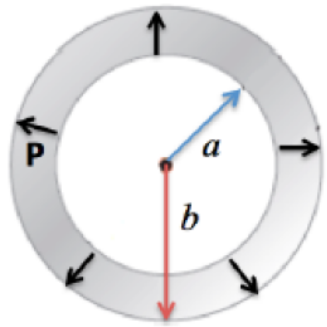




4. A thick spherical shell (inner radius 'a', outer radius 'b') is made of dielectric material with a "frozen-in" polarization

$$\boldsymbol{P} = \frac{k}{r}\hat{\boldsymbol{r}}$$

where 'k' is a constant and 'r' is the distance from the center. There is no free charge in the problem ($Q_f = 0$). Find the electric field in all three regions by two different methods:



(a) Locate all the bound charges, and use Gauss's law in terms of E,

$$\iint \boldsymbol{E}.\boldsymbol{dS} = \frac{1}{\epsilon_0}(Q_{enc}) = \frac{1}{\epsilon_0}(Q_{f,enc} + Q_{b,enc})$$

to calculate the field it produces. Here $Q_{f,enc}$ and $Q_{b,enc}$ denotes the enclosed free and bound charges respectively.

(b) Use Gauss's law in terms of **D**

$$\iint \boldsymbol{D}.\boldsymbol{dS} = Q_{f,enc}$$

to find **D**, and then obtain **E** using the relation

$$\boldsymbol{D} = \epsilon_0\boldsymbol{E} + \boldsymbol{P}$$